Route A Manuscript Attack

Expert inference validation report

Focus question
How do specialist results join without crossing finite, endpoint, or continuum boundaries?

Profile Id
route_a

Profile Name
Route A Manuscript Attack

Run Id
fixture.manuscript_result_elaboration.route_a

Target
manuscript:route-a

Trace Id
trace.3e1caf522e7f6153

Workflow Id
manuscript_result_elaboration

Workflow Name
Manuscript Result Elaboration

Workflow Version
1.0.0

Canonical packet SHA-256
67046d32dd47f0ee7a76ad2bfc2868284a6fb54f9fa7ed39f3c14e982a058f7f

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1 Expert reading contract

Public rationale boundary

Inspectable public rationale and evidence only. The inventory must not contain private chain-of-thought or derive scientific claims from layout, chronology, control authorization, or artifact integrity.

This document records inspectable premises, public warrants, evidence, conclusions, uncertainty, scope, authority, failures, and obligations. It does not expose private chain-of-thought. Execution authorization, artifact integrity, scientific epistemic status, and workflow disposition are independent. Chronological precedence does not mean logical or evidential support.

1.1 Status legend

Artifact integrity: whether referenced bytes and receipts validate **Scientific epistemic status:** the bounded support assigned to a claim **Workflow disposition:** what the workflow did with the item

2 Validation dashboard

Control coverage

1.0

Semantic coverage

1.0

Evidence resolution

1.0

Visible trace gaps

0

Required expert contrasts

["same-theory identity versus analogy", "symbolic check versus proof", "finite versus continuum boundary", "specialist output versus integrated result", "fork/join result versus common validation tail"]

3 Declared route and topology

Route spine

Specialist fork/join to common validation tail

Topology mode

fork_join

Step	Skill	Mode / observed state	Consumes	Produces
routea_attack	routea-paper-attack-orchestrator	required / observed	["PaperDraft"]	["RouteAAttackPlan"]
routea_open_points	routea-open-point-orchestrator	required / observed	["RouteAAttackPlan"]	["RouteAOpenPointPlan"]
routea_geometric_collar_builder	routea-geometric-collar-builder	specialist / observed	None	["ResultCandidatePacket"]
routea_same_theory_normalization_certifier	routea-same-theory-normalization-certifier	specialist / observed	None	["ResultCandidatePacket"]

Step	Skill	Mode / observed state	Consumes	Produces
routea_geometric_coefficient_extractor	routea-geometric-coefficient-extractor	specialist / observed	None	["ResultCandidatePacket"]
routea_finite_margin_evaluator	routea-finite-margin-evaluator	specialist / observed	None	["ResultCandidatePacket"]
routea_margin_endpoint_certifier	routea-margin-endpoint-certifier	specialist / observed	None	["ResultCandidatePacket"]
routea_endpoint_error_budget_certifier	routea-endpoint-error-budget-certifier	specialist / observed	None	["ResultCandidatePacket"]
routea_endpoint_local_integral_calculator	routea-endpoint-local-integral-calculator	specialist / observed	None	["ResultCandidatePacket"]
routea_same_theory_factorization_hunter	routea-same-theory-factorization-hunter	specialist / observed	None	["ResultCandidatePacket"]
routea_collar_redesign_forger	routea-collar-redesign-forger	specialist / observed	None	["ResultCandidatePacket"]
routea_continuum_gap_gatekeeper	routea-continuum-gap-gatekeeper	specialist / observed	None	["ResultCandidatePacket"]
routea_symbolic_audit	routea-symbolic-cas-auditor	conditional / observed	["ResultCandidatePacket"]	["SymbolicCASAudit"]
routea_integrate_result	routea-paper-result-integrator	required / observed	["ResultCandidatePacket"]	["RouteAIntegratedResult"]
routea_second_wave	routea-second-wave-manuscript-integrator	required / observed	["RouteAIntegratedResult"]	["IntegratedPaperDraft"]
certify_scope	results-theorem-scope-certifier	validator_only / observed	["RefinedResultCandidates"]	["TheoremScopeDecision"]
formalize_finite_object	results-finite-object-formalizer	conditional / observed	["RefinedResultCandidates"]	["FiniteObjectDefinition"]
derive_proof_packet	results-proof-packet-deriver	conditional / observed	["RefinedResultCandidates"]	["ResultsProofPacket"]
audit_prior_art	results-prior-art-disqualifier-auditor	validator_only / observed	["RefinedResultCandidates"]	["PriorArtBoundaryAudit"]
integrate_narrative	results-narrative-integrator	required / observed	["RefinedResultCandidates", "PriorArtBoundaryAudit"]	["IntegratedPaperDraft"]
validate_expansion	results-expansion-validator	validator_only / observed	["IntegratedPaperDraft"]	["ExpansionValidation"]

Step	Skill	Mode / observed state	Consumes	Produces
gate_readiness	results-paper-readiness-gatekeeper	validator_only / observed	["IntegratedPaperDraft", "ExpansionValidation"]	["PaperReadinessDecision"]

4 Profile topology and workflow-specific contrasts

Projection lens

manuscript_result_elaboration:route_a

Topology mode

fork_join

Required expert contrasts

["same-theory identity versus analogy", "symbolic check versus proof", "finite versus continuum boundary", "specialist output versus integrated result", "fork/join result versus common validation tail"]

Declared profile topology

{"specialist_after": "routea_open_points", "specialist_output": "ResultCandidatePacket", "specialist_step_ids": ["routea_geometric_collar_builder", "routea_same_theory_normalization_certifier", "routea_geometric_coefficient_extractor", "routea_finite_margin_evaluator", "routea_margin_endpoint_certifier", "routea_endpoint_error_budget_certifier", "routea_endpoint_local_integral_calculator", "routea_same_theory_factorization_hunter", "routea_collar_redesign_forger", "routea_continuum_gap_gatekeeper"], "validation_tail_adapter": {"from": "ResultCandidatePacket", "to": "RefinedResultCandidates"}, "validation_tail_start": "certify_scope"}

Observed specialist roles

["routea_geometric_collar_builder", "routea_same_theory_normalization_certifier", "routea_geometric_coefficient_extractor", "routea_finite_margin_evaluator", "routea_margin_endpoint_certifier", "routea_endpoint_error_budget_certifier", "routea_endpoint_local_integral_calculator", "routea_same_theory_factorization_hunter", "routea_collar_redesign_forger", "routea_continuum_gap_gatekeeper"]

A specialist fork or join is a workflow relation, not proof that specialist conclusions are mutually consistent. Finite, endpoint, asymptotic, and continuum boundaries remain separate unless an explicit validated inference connects them.

5 Expert inference dossiers

Each dossier is bound to its semantic event ID. Control-only events appear in the lifecycle appendix and are not scientific claims.

5.1 Inference 1: manuscript_result_elaboration.route_a.routea_attack.inference

---

Trace disposition

Complete public dossier

Step / skill

routea_attack / routea-paper-attack-orchestrator

Question

At routea_attack, how does PaperDraft bear on RouteAAttackPlan? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "PaperDraft: Independent specialist packets bind the same finite theory and declared endpoint. ", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At routea_attack, artifact_grounded_transformation checks PaperDraft before emitting RouteAAttackPlan. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea_attack produces RouteAAttackPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea_attack.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.5b6f606be7f186

Alternative / falsifier / failure links

alternative.99a32ecdf811b2, falsifier.4eec699600e8d0

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea_attack-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.2 Inference 2: manuscript_result_elaboration.route_a.routea_open_points.inference

---

Trace disposition

Complete public dossier

Step / skill

routea_open_points / routea-open-point-orchestrator

Question

At routea_open_points, how does RouteAAttackPlan bear on RouteAOpenPointPlan? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAAttackPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At routea_open_points, artifact_grounded_transformation checks RouteAAttackPlan before emitting RouteAOpenPointPlan. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea_open_points produces RouteAOpenPointPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea_open_points.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.315ac9b9630548

Alternative / falsifier / failure links

alternative.d152978badc48d, falsifier.07e3e2f3514d79

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea_open_points-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.3 Inference 3: manuscript_result_elaboration.route_a.routea_symbolic_audit.inference

Trace disposition

Complete public dossier

Step / skill

routea_symbolic_audit / routea-symbolic-cas-auditor

Question

At routea_symbolic_audit, how does ResultCandidatePacket bear on SymbolicCASAudit? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "ResultCandidatePacket: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_symbolic_identity

Bounded public warrant

At routea_symbolic_audit, verify_symbolic_identity checks ResultCandidatePacket before emitting SymbolicCASAudit. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea_symbolic_audit produces SymbolicCASAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea_symbolic_audit.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.15ddea226fb640

Alternative / falsifier / failure links

alternative.114eb8b2d14d29, falsifier.b13f8b2e2bf9b7

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea_symbolic_audit-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.4 Inference 4: manuscript_result_elaboration.route_a.routea_integrate_result.inference

---

Trace disposition

Complete public dossier

Step / skill

routea_integrate_result / routea-paper-result-integrator

Question

At routea_integrate_result, how does ResultCandidatePacket bear on RouteAIntegratedResult? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "ResultCandidatePacket: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

apply_traceable_artifact_delta

Bounded public warrant

At routea_integrate_result, apply_traceable_artifact_delta checks ResultCandidatePacket before emitting RouteAIntegratedResult. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea_integrate_result produces RouteAIntegratedResult: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea_integrate_result.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.4038a046d42c12

Alternative / falsifier / failure links

alternative.83089fc60db716, falsifier.835de479c180bf

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea_integrate_result-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.5 Inference 5: manuscript_result_elaboration.route_a.routea_second_wave.inference

---

Trace disposition

Complete public dossier

Step / skill

routea_second_wave / routea-second-wave-manuscript-integrator

Question

At routea_second_wave, how does RouteAIntegratedResult bear on IntegratedPaperDraft? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAIntegratedResult: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At routea_second_wave, artifact_grounded_transformation checks RouteAIntegratedResult before emitting IntegratedPaperDraft. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea_second_wave produces IntegratedPaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea_second_wave.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.f864c300875269

Alternative / falsifier / failure links

alternative.1d7f3776c1b400, falsifier.39c9ce6d29f4a2

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea_second_wave-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.6 Inference 6: manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-geometric-collar-builder / routea-geometric-collar-builder

Question

At routea-geometric-collar-builder, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_bounded_specialist_obligation

Bounded public warrant

At routea-geometric-collar-builder, verify_bounded_specialist_obligation checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-geometric-collar-builder produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.8f455a822ffefc

Alternative / falsifier / failure links

alternative.3f77689193b02b, falsifier.800db42b415415

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-geometric-collar-builder-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.7 Inference 7: manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-same-theory-normalization-certifier / routea-same-theory-normalization-certifier

Question

At routea-same-theory-normalization-certifier, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At routea-same-theory-normalization-certifier, artifact_grounded_transformation checks RouteAOpen PointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-same-theory-normalization-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.89afe62c827cf9

Alternative / falsifier / failure links

alternative.1e9d284690ba99, falsifier.3c99a2a7ce3bb4

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-same-theory-normalization-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.8 Inference 8: manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-geometric-coefficient-extractor / routea-geometric-coefficient-extractor

Question

At routea-geometric-coefficient-extractor, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At routea-geometric-coefficient-extractor, artifact_grounded_transformation checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-geometric-coefficient-extractor produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.e57a22c1a0a375

Alternative / falsifier / failure links

alternative.54a928154bf647, falsifier.6f51ddd6a8f986

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-geometric-coefficient-extractor-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.9 Inference 9: manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference

Trace disposition

Complete public dossier

Step / skill

routea-finite-margin-evaluator / routea-finite-margin-evaluator

Question

At routea-finite-margin-evaluator, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_bounded_specialist_obligation

Bounded public warrant

At routea-finite-margin-evaluator, verify_bounded_specialist_obligation checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-finite-margin-evaluator produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.2fc65723cf2e10

Alternative / falsifier / failure links

alternative.d94eb4a71d5769, falsifier.70e1675ca7e9c8

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-finite-margin-evaluator-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.10 Inference 10: manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-margin-endpoint-certifier / routea-margin-endpoint-certifier

Question

At routea-margin-endpoint-certifier, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_bounded_specialist_obligation

Bounded public warrant

At routea-margin-endpoint-certifier, verify_bounded_specialist_obligation checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-margin-endpoint-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.1c29564458a55f

Alternative / falsifier / failure links

alternative.82f5d1b20c7c7f, falsifier.7732eed567cc0d

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-margin-endpoint-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.11 Inference 11: manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-endpoint-error-budget-certifier / routea-endpoint-error-budget-certifier

Question

At routea-endpoint-error-budget-certifier, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_bounded_specialist_obligation

Bounded public warrant

At routea-endpoint-error-budget-certifier, verify_bounded_specialist_obligation checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-endpoint-error-budget-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.4fac32c287b49e

Alternative / falsifier / failure links

alternative.1673ce90ffa7b8, falsifier.5efd3cd8561ccc

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-endpoint-error-budget-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.12 Inference 12: manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-same-theory-factorization-hunter / routea-same-theory-factorization-hunter

Question

At routea-same-theory-factorization-hunter, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

verify_bounded_specialist_obligation

Bounded public warrant

At routea-same-theory-factorization-hunter, verify_bounded_specialist_obligation checks RouteAOpen PointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-same-theory-factorization-hunter produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.fe49659fa93aea

Alternative / falsifier / failure links

alternative.a5f38e52cb4bfa, falsifier.79e62b2f313a37

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-same-theory-factorization-hunter-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.13 Inference 13: manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-collar-redesign-forger / routea-collar-redesign-forger

Question

At routea-collar-redesign-forger, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

generate_independent_alternatives

Bounded public warrant

At routea-collar-redesign-forger, generate_independent_alternatives checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-collar-redesign-forger produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.795cc6fc69f022

Alternative / falsifier / failure links

alternative.1ddbd74c9dee69, falsifier.2ebe7eeca11266

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-collar-redesign-forger-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.14 Inference 14: manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference

---

Trace disposition

Complete public dossier

Step / skill

routea-continuum-gap-gatekeeper / routea-continuum-gap-gatekeeper

Question

At routea-continuum-gap-gatekeeper, how does RouteAOpenPointPlan bear on ResultCandidatePacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RouteAOpenPointPlan: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

adversarially_test_or_split

Bounded public warrant

At routea-continuum-gap-gatekeeper, adversarially_test_or_split checks RouteAOpenPointPlan before emitting ResultCandidatePacket. Authority boundary: May integrate a bounded Route-A result into the parent manuscript workflow; may not promote a finite packet past the common validation tail.

Conclusion

routea-continuum-gap-gatekeeper produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.76f139d78fc9ef

Alternative / falsifier / failure links

alternative.70ce25c5a2978b, falsifier.19281f30f21cf6

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["routea-continuum-gap-gatekeeper-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.15 Inference 15: manuscript_result_elaboration.route_a.certify_scope.inference

Trace disposition

Complete public dossier

Step / skill

certify_scope / results-theorem-scope-certifier

Question

At certify_scope, how does RefinedResultCandidates bear on TheoremScopeDecision? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RefinedResultCandidates: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

bind_claim_scope

Bounded public warrant

At certify_scope, bind_claim_scope checks RefinedResultCandidates before emitting TheoremScopeDecision . Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

certify_scope produces TheoremScopeDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.certify_scope.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.29ba0e221db096

Alternative / falsifier / failure links

alternative.ed1ffe23834508, falsifier.9a422bcce0bc75

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["certify_scope-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.16 Inference 16: manuscript_result_elaboration.route_a.formalize_finite_object.inference

---

Trace disposition

Complete public dossier

Step / skill

formalize_finite_object / results-finite-object-formalizer

Question

At formalize_finite_object, how does RefinedResultCandidates bear on FiniteObjectDefinition? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RefinedResultCandidates: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

derive_formal_obligations

Bounded public warrant

At formalize_finite_object, derive_formal_obligations checks RefinedResultCandidates before emitting FiniteObjectDefinition. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

formalize_finite_object produces FiniteObjectDefinition: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.formalize_finite_object.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.ffd73b47728ad7

Alternative / falsifier / failure links

alternative.c159a2b011a916, falsifier.69b5f4b273169f

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["formalize_finite_object-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.17 Inference 17: manuscript_result_elaboration.route_a.derive_proof_packet.inference

---

Trace disposition

Complete public dossier

Step / skill

derive_proof_packet / results-proof-packet-deriver

Question

At derive_proof_packet, how does RefinedResultCandidates bear on ResultsProofPacket? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RefinedResultCandidates: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

validate_exact_claim_boundary

Bounded public warrant

At derive_proof_packet, validate_exact_claim_boundary checks RefinedResultCandidates before emitting ResultsProofPacket. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

derive_proof_packet produces ResultsProofPacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.derive_proof_packet.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.deb4ef68f414b5

Alternative / falsifier / failure links

alternative.417c1bb6ce8ae6, falsifier.1fa6b256b8a0e9

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["derive_proof_packet-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.18 Inference 18: manuscript_result_elaboration.route_a.audit_prior_art.inference

---

Trace disposition

Complete public dossier

Step / skill

audit_prior_art / results-prior-art-disqualifier-auditor

Question

At audit_prior_art, how does RefinedResultCandidates bear on PriorArtBoundaryAudit? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RefinedResultCandidates: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

test_prior_art_disqualifiers

Bounded public warrant

At audit_prior_art, test_prior_art_disqualifiers checks RefinedResultCandidates before emitting PriorArtBoundaryAudit. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

audit_prior_art produces PriorArtBoundaryAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.audit_prior_art.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.7807d1cab61465

Alternative / falsifier / failure links

alternative.11b1fc3a8ceadf, falsifier.e9030e779e0775

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["audit_prior_art-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.19 Inference 19: manuscript_result_elaboration.route_a.integrate_narrative.inference

---

Trace disposition

Complete public dossier

Step / skill

integrate_narrative / results-narrative-integrator

Question

At integrate_narrative, how does RefinedResultCandidates bear on IntegratedPaperDraft? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "RefinedResultCandidates: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

apply_traceable_artifact_delta

Bounded public warrant

At integrate_narrative, apply_traceable_artifact_delta checks RefinedResultCandidates before emitting IntegratedPaperDraft. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

integrate_narrative produces IntegratedPaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.integrate_narrative.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.a8bb26ed9ec475

Alternative / falsifier / failure links

alternative.4bddb7bef1d62f, falsifier.e9392c6f59fc5e

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["integrate_narrative-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.20 Inference 20: manuscript_result_elaboration.route_a.validate_expansion.inference

---

Trace disposition

Complete public dossier

Step / skill

validate_expansion / results-expansion-validator

Question

At validate_expansion, how does IntegratedPaperDraft bear on ExpansionValidation? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "IntegratedPaperDraft: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At validate_expansion, artifact_grounded_transformation checks IntegratedPaperDraft before emitting ExpansionValidation. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

validate_expansion produces ExpansionValidation: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.validate_expansion.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.8da5549746a255

Alternative / falsifier / failure links

alternative.5f91b2855aa586, falsifier.aebb2cb619c019

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["validate_expansion-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.21 Inference 21: manuscript_result_elaboration.route_a.gate_readiness.inference

Trace disposition

Complete public dossier

Step / skill

gate_readiness / results-paper-readiness-gatekeeper

Question

At gate_readiness, how does IntegratedPaperDraft bear on PaperReadinessDecision? Do the specialist packets establish the Route-A identity, endpoint, and finite margin?

Typed premises

[{"statement": "IntegratedPaperDraft: Independent specialist packets bind the same finite theory and declared endpoint.", "type": "artifact"}]

Transformation / operator

validate_exact_claim_boundary

Bounded public warrant

At gate_readiness, validate_exact_claim_boundary checks IntegratedPaperDraft before emitting PaperReadinessDecision. Authority boundary: May elaborate and integrate supported manuscript claims; may not perform implicit discovery, erase disqualifiers, or promote finite evidence to continuum scope.

Conclusion

gate_readiness produces PaperReadinessDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}

Causal parents

["manuscript_result_elaboration.route_a.gate_readiness.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.4f120a9286f066

Alternative / falsifier / failure links

alternative.e67efabfdbd994, falsifier.4754b6261e62e6

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["gate_readiness-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

6 Alternatives, falsifiers, and failed paths

ID	Kind	Inference	Description	Disposition
alternative.99a32ecdf811b2	alternative	manuscript_result_elaboration.route_a.routea_attack.inference	Do not emit RouteAAttackPlan: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.4eec699600e8d0	falsifier	manuscript_result_elaboration.route_a.routea_attack.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.d152978badc48d	alternative	manuscript_result_elaboration.route_a.routea_open_points.inference	Do not emit RouteAOpenPointPlan: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.07e3e2f3514d79	falsifier	manuscript_result_elaboration.route_a.routea_open_points.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.114eb8b2d14d29	alternative	manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	Do not emit SymbolicCASAudit: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.b13f8b2e2bf9b7	falsifier	manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.83089fc60db716	alternative	manuscript_result_elaboration.route_a.routea_integrate_result.inference	Do not emit RouteAIntegratedResult: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.835de479c180bf	falsifier	manuscript_result_elaboration.route_a.routea_integrate_result.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.1d7f3776c1b400	alternative	manuscript_result_elaboration.route_a.routea_second_wave.inference	Do not emit IntegratedPaperDraft: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.39c9ce6d29f4a2	falsifier	manuscript_result_elaboration.route_a.routea_second_wave.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.3f77689193b02b	alternative	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.800db42b415415	falsifier	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.1e9d284690ba99	alternative	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.3c99a2a7ce3bb4	falsifier	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.54a928154bf647	alternative	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.6f51ddd6a8f986	falsifier	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.d94eb4a71d5769	alternative	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.70e1675ca7e9c8	falsifier	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.82f5d1b20c7c7f	alternative	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.7732eed567cc0d	falsifier	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded

ID	Kind	Inference	Description	Disposition
alternative.1673ce90ffa7b8	alternative	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.5efd3cd8561ccc	falsifier	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.1ffd9275dd61fe	alternative	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	accepted
falsifier.1838f8df815cea	falsifier	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternatives_and_failure.760021ef51d7242e	failure	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	routea-endpoint-local-integral-calculator rejects ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	Not recorded
alternative.a5f38e52cb4bfa	alternative	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.79e62b2f313a37	falsifier	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.1ddbd74c9dee69	alternative	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.2ebe7eeca11266	falsifier	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.70ce25c5a2978b	alternative	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	Do not emit ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.19281f30f21cf6	falsifier	manuscript_result_elaboration.route_a.route-a-continuum-gap-gatekeeper.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.ed1ffe23834508	alternative	manuscript_result_elaboration.route_a.certify_scope.inference	Do not emit TheoremScopeDecision: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.9a422bcce0bc75	falsifier	manuscript_result_elaboration.route_a.certify_scope.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.c159a2b011a916	alternative	manuscript_result_elaboration.route_a.formalize_finite_object.inference	Do not emit FiniteObjectDefinition: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.69b5f4b273169f	falsifier	manuscript_result_elaboration.route_a.formalize_finite_object.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.417c1bb6ce8ae6	alternative	manuscript_result_elaboration.route_a.derive_proof_packet.inference	Do not emit ResultsProofPacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.1fa6b256b8a0e9	falsifier	manuscript_result_elaboration.route_a.derive_proof_packet.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.11b1fc3a8ceadf	alternative	manuscript_result_elaboration.route_a.audit_prior_art.inference	Do not emit PriorArtBoundaryAudit: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.e9030e779e0775	falsifier	manuscript_result_elaboration.route_a.audit_prior_art.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.4bddb7bef1d62f	alternative	manuscript_result_elaboration.route_a.integrate_narrative.inference	Do not emit IntegratedPaperDraft: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.e9392c6f59fc5e	falsifier	manuscript_result_elaboration.route_a.integrate_narrative.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.5f91b2855aa586	alternative	manuscript_result_elaboration.route_a.validate_expansion.inference	Do not emit ExpansionValidation: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.aebb2cb619c019	falsifier	manuscript_result_elaboration.route_a.validate_expansion.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
alternative.e67efabfdbd994	alternative	manuscript_result_elaboration.route_a.gate_readiness.inference	Do not emit PaperReadinessDecision: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	rejected
falsifier.4754b6261e62e6	falsifier	manuscript_result_elaboration.route_a.gate_readiness.inference	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded

7 Cross-cutting validation ledgers

7.1 Scope transitions

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea_attack.inference	{"event_id": "manuscript_result_elaboration.route_a.routea_attack.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea_attack"}
manuscript_result_elaboration.route_a.routea_open_points.inference	{"event_id": "manuscript_result_elaboration.route_a.routea_open_points.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea_open_points"}
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	{"event_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea_symbolic_audit"}
manuscript_result_elaboration.route_a.routea_integrate_result.inference	{"event_id": "manuscript_result_elaboration.route_a.routea_integrate_result.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea_integrate_result"}
manuscript_result_elaboration.route_a.routea_second_wave.inference	{"event_id": "manuscript_result_elaboration.route_a.routea_second_wave.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea_second_wave"}
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-geometric-collar-builder"}
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-same-theory-normalization-certifier"}
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-geometric-coefficient-extractor"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-finite-margin-evaluator"}
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-margin-endpoint-certifier"}
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-endpoint-error-budget-certifier"}
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-endpoint-local-integral-calculator"}
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-same-theory-factorization-hunter"}
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-collar-redesign-forger"}
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	{"event_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "routea-continuum-gap-gatekeeper"}
manuscript_result_elaboration.route_a.certify_scope.inference	{"event_id": "manuscript_result_elaboration.route_a.certify_scope.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "certify_scope"}
manuscript_result_elaboration.route_a.formalize_finite_object.inference	{"event_id": "manuscript_result_elaboration.route_a.formalize_finite_object.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "formalize_finite_object"}
manuscript_result_elaboration.route_a.derive_proof_packet.inference	{"event_id": "manuscript_result_elaboration.route_a.derive_proof_packet.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "derive_proof_packet"}
manuscript_result_elaboration.route_a.audit_prior_art.inference	{"event_id": "manuscript_result_elaboration.route_a.audit_prior_art.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "audit_prior_art"}
manuscript_result_elaboration.route_a.integrate_narrative.inference	{"event_id": "manuscript_result_elaboration.route_a.integrate_narrative.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "integrate_narrative"}
manuscript_result_elaboration.route_a.validate_expansion.inference	{"event_id": "manuscript_result_elaboration.route_a.validate_expansion.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "validate_expansion"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.gate_readiness.inference	{"event_id": "manuscript_result_elaboration.route_a.gate_readiness.inference", "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "step_id": "gate_readiness"}

7.2 Authority boundaries

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea_attack.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea_attack.inference", "exercised": ["routea_attack-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea_attack"}
manuscript_result_elaboration.route_a.routea_open_points.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea_open_points.inference", "exercised": ["routea_open_points-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea_open_points"}
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.inference", "exercised": ["routea_symbolic_audit-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea_symbolic_audit"}
manuscript_result_elaboration.route_a.routea_integrate_result.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea_integrate_result.inference", "exercised": ["routea_integrate_result-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea_integrate_result"}
manuscript_result_elaboration.route_a.routea_second_wave.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea_second_wave.inference", "exercised": ["routea_second_wave-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea_second_wave"}
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference", "exercised": ["routea-geometric-collar-builder-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-geometric-collar-builder"}
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference", "exercised": ["routea-same-theory-normalization-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-same-theory-normalization-certifier"}
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference", "exercised": ["routea-geometric-coefficient-extractor-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-geometric-coefficient-extractor"}
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference", "exercised": ["routea-finite-margin-evaluator-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-finite-margin-evaluator"}
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference", "exercised": ["routea-margin-endpoint-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-margin-endpoint-certifier"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference", "exercised": ["routea-endpoint-error-budget-certifier-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-endpoint-error-budget-certifier"}
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "exercised": ["routea-endpoint-local-integral-calculator-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-endpoint-local-integral-calculator"}
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference", "exercised": ["routea-same-theory-factorization-hunter-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-same-theory-factorization-hunter"}
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference", "exercised": ["routea-collar-redesign-forger-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-collar-redesign-forger"}
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference", "exercised": ["routea-continuum-gap-gatekeeper-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "routea-continuum-gap-gatekeeper"}
manuscript_result_elaboration.route_a.certify_scope.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.certify_scope.inference", "exercised": ["certify_scope-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "certify_scope"}
manuscript_result_elaboration.route_a.formalize_finite_object.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.formalize_finite_object.inference", "exercised": ["formalize_finite_object-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "formalize_finite_object"}
manuscript_result_elaboration.route_a.derive_proof_packet.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.derive_proof_packet.inference", "exercised": ["derive_proof_packet-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "derive_proof_packet"}
manuscript_result_elaboration.route_a.audit_prior_art.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.audit_prior_art.inference", "exercised": ["audit_prior_art-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "audit_prior_art"}
manuscript_result_elaboration.route_a.integrate_narrative.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.integrate_narrative.inference", "exercised": ["integrate_narrative-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "integrate_narrative"}
manuscript_result_elaboration.route_a.validate_expansion.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.validate_expansion.inference", "exercised": ["validate_expansion-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "validate_expansion"}
manuscript_result_elaboration.route_a.gate_readiness.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "manuscript_result_elaboration.route_a.gate_readiness.inference", "exercised": ["gate_readiness-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "gate_readiness"}

7.3 Open and discharged obligations

Semantic ID	Complete recorded detail
obligation.0be0f054580524	{"event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "obligation_id": "obligation.0be0f054580524", "requirement": "A negative certified margin or same-theory mismatch rejects the integrated result.", "status": "open"}

7.4 Handoffs

Semantic ID	Complete recorded detail
manuscript_result_elaboration.handoff	{"accepted": true, "destination_workflow_id": "theorem_proof_delivery", "event_id": "manuscript_result_elaboration.handoff"}

A Complete chronological lifecycle ledger

This ledger records execution order. A precedes relation is not evidential support or entailment.

Event ID	Seq.	Family	Step	Disposition	Parents
manuscript_result_elaboration.route_a.routea_attack.control	1	authorization	routea_attack	authorized	None
manuscript_result_elaboration.route_a.routea_attack.inference	2	inference	routea_attack	accepted	["manuscript_result_elaboration.route_a.routea_attack.control"]
manuscript_result_elaboration.route_a.routea_open_points.control	3	authorization	routea_open_points	authorized	["manuscript_result_elaboration.route_a.routea_attack.inference"]
manuscript_result_elaboration.route_a.routea_open_points.inference	4	inference	routea_open_points	accepted	["manuscript_result_elaboration.route_a.routea_open_points.control"]
manuscript_result_elaboration.route_a.routea_symbolic_audit.control	5	authorization	routea_symbolic_audit	authorized	["manuscript_result_elaboration.route_a.routea_open_points.inference"]
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	6	inference	routea_symbolic_audit	accepted	["manuscript_result_elaboration.route_a.routea_symbolic_audit.control"]
manuscript_result_elaboration.route_a.routea_integrate_result.control	7	authorization	routea_integrate_result	authorized	["manuscript_result_elaboration.route_a.routea_symbolic_audit.inference"]
manuscript_result_elaboration.route_a.routea_integrate_result.inference	8	inference	routea_integrate_result	accepted	["manuscript_result_elaboration.route_a.routea_integrate_result.control"]
manuscript_result_elaboration.route_a.routea_second_wave.control	9	authorization	routea_second_wave	authorized	["manuscript_result_elaboration.route_a.routea_integrate_result.inference"]

Event ID	Seq.	Family	Step	Disposition	Parents
manuscript_result_elaboration.route_a.routea_second_wave.inference	10	inference	routea_second_wave	accepted	["manuscript_result_elaboration.route_a.routea_second_wave.control"]
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	11	authorization	routea-geometric-collar-builder	authorized	["manuscript_result_elaboration.route_a.routea_second_wave.inference"]
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	12	inference	routea-geometric-collar-builder	accepted	["manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control"]
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	13	authorization	routea-same-theory-normalization-certifier	authorized	["manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference"]
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	14	inference	routea-same-theory-normalization-certifier	accepted	["manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control"]
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	15	authorization	routea-geometric-coefficient-extractor	authorized	["manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference"]
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	16	inference	routea-geometric-coefficient-extractor	accepted	["manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control"]
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	17	authorization	routea-finite-margin-evaluator	authorized	["manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference"]
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	18	inference	routea-finite-margin-evaluator	accepted	["manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control"]
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	19	authorization	routea-margin-endpoint-certifier	authorized	["manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference"]
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	20	inference	routea-margin-endpoint-certifier	accepted	["manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control"]
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	21	authorization	routea-endpoint-error-budget-certifier	authorized	["manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference"]

Event ID	Seq.	Family	Step	Disposition	Parents
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	22	inference	routea-endpoint-error-budget-certifier	accepted	["manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control"]
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	23	authorization	routea-endpoint-local-integral-calculator	authorized	["manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference"]
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	24	failure	routea-endpoint-local-integral-calculator	failed	["manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control"]
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	25	authorization	routea-same-theory-factorization-hunter	authorized	["manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference"]
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	26	inference	routea-same-theory-factorization-hunter	accepted	["manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control"]
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	27	authorization	routea-collar-redesign-forger	authorized	["manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference"]
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	28	inference	routea-collar-redesign-forger	accepted	["manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control"]
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	29	authorization	routea-continuum-gap-gatekeeper	authorized	["manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference"]
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	30	inference	routea-continuum-gap-gatekeeper	accepted	["manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control"]
manuscript_result_elaboration.route_a.certify_scope.control	31	authorization	certify_scope	authorized	["manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference"]
manuscript_result_elaboration.route_a.certify_scope.inference	32	inference	certify_scope	accepted	["manuscript_result_elaboration.route_a.certify_scope.control"]
manuscript_result_elaboration.route_a.formalize_finite_object.control	33	authorization	formalize_finite_object	authorized	["manuscript_result_elaboration.route_a.certify_scope.inference"]
manuscript_result_elaboration.route_a.formalize_finite_object.inference	34	inference	formalize_finite_object	accepted	["manuscript_result_elaboration.route_a.formalize_finite_object.control"]

Event ID	Seq.	Family	Step	Disposition	Parents
manuscript_result_elaboration.route_a.derive_proof_packet.control	35	authorization	derive_proof_packet	authorized	["manuscript_result_elaboration.route_a.formalize_finite_object.inference"]
manuscript_result_elaboration.route_a.derive_proof_packet.inference	36	inference	derive_proof_packet	accepted	["manuscript_result_elaboration.route_a.derive_proof_packet.control"]
manuscript_result_elaboration.route_a.audit_prior_art.control	37	authorization	audit_prior_art	authorized	["manuscript_result_elaboration.route_a.derive_proof_packet.inference"]
manuscript_result_elaboration.route_a.audit_prior_art.inference	38	inference	audit_prior_art	accepted	["manuscript_result_elaboration.route_a.audit_prior_art.control"]
manuscript_result_elaboration.route_a.integrate_narrative.control	39	authorization	integrate_narrative	authorized	["manuscript_result_elaboration.route_a.audit_prior_art.inference"]
manuscript_result_elaboration.route_a.integrate_narrative.inference	40	inference	integrate_narrative	accepted	["manuscript_result_elaboration.route_a.integrate_narrative.control"]
manuscript_result_elaboration.route_a.validate_expansion.control	41	authorization	validate_expansion	authorized	["manuscript_result_elaboration.route_a.integrate_narrative.inference"]
manuscript_result_elaboration.route_a.validate_expansion.inference	42	inference	validate_expansion	accepted	["manuscript_result_elaboration.route_a.validate_expansion.control"]
manuscript_result_elaboration.route_a.gate_readiness.control	43	authorization	gate_readiness	authorized	["manuscript_result_elaboration.route_a.validate_expansion.inference"]
manuscript_result_elaboration.route_a.gate_readiness.inference	44	inference	gate_readiness	accepted	["manuscript_result_elaboration.route_a.gate_readiness.control"]
manuscript_result_elaboration.handoff	45	handoff	Not recorded	handed_off	["manuscript_result_elaboration.route_a.gate_readiness.inference"]

B Claim ledger

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea_attack.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea_attack.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea_attack.inference", "evidence_ids": ["evidence.5b6f606be7f186"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea_attack produces RouteAAttackPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea_open_points.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea_open_points.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea_open_points.inference", "evidence_ids": ["evidence.315ac9b9630548"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea_open_points produces RouteAOpenPointPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.inference", "evidence_ids": ["evidence.15ddea226fb640"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea_symbolic_audit produces SymbolicCASAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea_integrate_result.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea_integrate_result.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea_integrate_result.inference", "evidence_ids": ["evidence.4038a046d42c12"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea_integrate_result produces RouteAIntegratedResult: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea_second_wave.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea_second_wave.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea_second_wave.inference", "evidence_ids": ["evidence.f864c300875269"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea_second_wave produces IntegratedPaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference", "evidence_ids": ["evidence.8f455a822ffefc"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-geometric-collar-builder produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference", "evidence_ids": ["evidence.89afe62c827cf9"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-same-theory-normalization-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference", "evidence_ids": ["evidence.e57a22c1a0a375"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-geometric-coefficient-extractor produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference", "evidence_ids": ["evidence.2fc65723cf2e10"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-finite-margin-evaluator produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference", "evidence_ids": ["evidence.1c29564458a55f"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-margin-endpoint-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference", "evidence_ids": ["evidence.4fac32c287b49e"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-endpoint-error-budget-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "epistemic_status": "quarantined", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "evidence_ids": ["evidence.1952ee704db1f8"], "integrity_status": "integrity_valid", "obligation_ids": ["obligation.0be0f054580524"], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-endpoint-local-integral-calculator rejects ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.", "trace_incomplete": false, "workflow_disposition": "failed"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference", "evidence_ids": ["evidence.fe49659fa93aea"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-same-theory-factorization-hunter produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference", "evidence_ids": ["evidence.795cc6fc69f022"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-collar-redesign-forger produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	{"claim_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference", "evidence_ids": ["evidence.76f139d78fc9ef"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "routea-continuum-gap-gatekeeper produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.certify_scope.inference	{"claim_id": "manuscript_result_elaboration.route_a.certify_scope.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.certify_scope.inference", "evidence_ids": ["evidence.29ba0e221db096"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "certify_scope produces TheoremScopeDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.formalize_finite_object.inference	{"claim_id": "manuscript_result_elaboration.route_a.formalize_finite_object.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.formalize_finite_object.inference", "evidence_ids": ["evidence.ffd73b47728ad7"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "formalize_finite_object produces FiniteObjectDefinition: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.derive_proof_packet.inference	{"claim_id": "manuscript_result_elaboration.route_a.derive_proof_packet.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.derive_proof_packet.inference", "evidence_ids": ["evidence.deb4ef68f414b5"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "derive_proof_packet produces ResultsProofPacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.audit_prior_art.inference	{"claim_id": "manuscript_result_elaboration.route_a.audit_prior_art.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.audit_prior_art.inference", "evidence_ids": ["evidence.7807d1cab61465"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "audit_prior_art produces PriorArtBoundaryAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.integrate_narrative.inference	{"claim_id": "manuscript_result_elaboration.route_a.integrate_narrative.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.integrate_narrative.inference", "evidence_ids": ["evidence.a8bb26ed9ec475"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "integrate_narrative produces IntegratedPaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.validate_expansion.inference	{"claim_id": "manuscript_result_elaboration.route_a.validate_expansion.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.validate_expansion.inference", "evidence_ids": ["evidence.8da5549746a255"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "validate_expansion produces ExpansionValidation: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}
manuscript_result_elaboration.route_a.gate_readiness.inference	{"claim_id": "manuscript_result_elaboration.route_a.gate_readiness.inference", "epistemic_status": "evidence_backed", "event_id": "manuscript_result_elaboration.route_a.gate_readiness.inference", "evidence_ids": ["evidence.4f120a9286f066"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "bounded result at validation tail", "before": "specialist fan-out", "model": "same-theory endpoint packet", "regime": "finite Route-A"}, "text": "gate_readiness produces PaperReadinessDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.", "trace_incomplete": false, "workflow_disposition": "accepted"}

C Typed evidence ledger

Semantic ID	Complete recorded detail
evidence.5b6f606be7f186	{"artifact_type": "PaperDraft", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea_attack.inference", "evidence_id": "evidence.5b6f606be7f186", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.315ac9b9630548	{"artifact_type": "RouteAAttackPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea_open_points.inference", "evidence_id": "evidence.315ac9b9630548", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.15ddea226fb640	{"artifact_type": "ResultCandidatePacket", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.inference", "evidence_id": "evidence.15ddea226fb640", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

Semantic ID	Complete recorded detail
evidence.4038a046d42c12	{"artifact_type": "ResultCandidatePacket", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea_integrate_result.inference", "evidence_id": "evidence.4038a046d42c12", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.f864c300875269	{"artifact_type": "RouteAIntegratedResult", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea_second_wave.inference", "evidence_id": "evidence.f864c300875269", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.8f455a822ffefc	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference", "evidence_id": "evidence.8f455a822ffefc", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.89afe62c827cf9	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference", "evidence_id": "evidence.89afe62c827cf9", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.e57a22c1a0a375	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference", "evidence_id": "evidence.e57a22c1a0a375", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.2fc65723cf2e10	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference", "evidence_id": "evidence.2fc65723cf2e10", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.1c29564458a55f	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference", "evidence_id": "evidence.1c29564458a55f", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.4fac32c287b49e	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference", "evidence_id": "evidence.4fac32c287b49e", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.1952ee704db1f8	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference", "evidence_id": "evidence.1952ee704db1f8", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "refutes"}
evidence.fe49659fa93aea	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference", "evidence_id": "evidence.fe49659fa93aea", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.795cc6fc69f022	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference", "evidence_id": "evidence.795cc6fc69f022", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.76f139d78fc9ef	{"artifact_type": "RouteAOpenPointPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference", "evidence_id": "evidence.76f139d78fc9ef", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.29ba0e221db096	{"artifact_type": "RefinedResultCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.certify_scope.inference", "evidence_id": "evidence.29ba0e221db096", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

Semantic ID	Complete recorded detail
evidence.ffd73b47728ad7	{"artifact_type": "RefinedResultCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.formalize_finite_object.inference", "evidence_id": "evidence.ffd73b47728ad7", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.deb4ef68f414b5	{"artifact_type": "RefinedResultCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.derive_proof_packet.inference", "evidence_id": "evidence.deb4ef68f414b5", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.7807d1cab61465	{"artifact_type": "RefinedResultCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.audit_prior_art.inference", "evidence_id": "evidence.7807d1cab61465", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.a8bb26ed9ec475	{"artifact_type": "RefinedResultCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.integrate_narrative.inference", "evidence_id": "evidence.a8bb26ed9ec475", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.8da5549746a255	{"artifact_type": "IntegratedPaperDraft", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.validate_expansion.inference", "evidence_id": "evidence.8da5549746a255", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.4f120a9286f066	{"artifact_type": "IntegratedPaperDraft", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "manuscript_result_elaboration.route_a.gate_readiness.inference", "evidence_id": "evidence.4f120a9286f066", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

D Decision and gate ledger

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea_attack.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea_attack.control", "family": "authorization", "reason": "", "step_id": "routea_attack"}
manuscript_result_elaboration.route_a.routea_open_points.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea_open_points.control", "family": "authorization", "reason": "", "step_id": "routea_open_points"}
manuscript_result_elaboration.route_a.routea_symbolic_audit.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea_symbolic_audit.control", "family": "authorization", "reason": "", "step_id": "routea_symbolic_audit"}
manuscript_result_elaboration.route_a.routea_integrate_result.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea_integrate_result.control", "family": "authorization", "reason": "", "step_id": "routea_integrate_result"}
manuscript_result_elaboration.route_a.routea_second_wave.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea_second_wave.control", "family": "authorization", "reason": "", "step_id": "routea_second_wave"}
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control", "family": "authorization", "reason": "", "step_id": "routea-geometric-collar-builder"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control", "family": "authorization", "reason": "", "step_id": "routea-same-theory-normalization-certifier"}
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control", "family": "authorization", "reason": "", "step_id": "routea-geometric-coefficient-extractor"}
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control", "family": "authorization", "reason": "", "step_id": "routea-finite-margin-evaluator"}
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control", "family": "authorization", "reason": "", "step_id": "routea-margin-endpoint-certifier"}
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control", "family": "authorization", "reason": "", "step_id": "routea-endpoint-error-budget-certifier"}
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control", "family": "authorization", "reason": "", "step_id": "routea-endpoint-local-integral-calculator"}
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control", "family": "authorization", "reason": "", "step_id": "routea-same-theory-factorization-hunter"}
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control", "family": "authorization", "reason": "", "step_id": "routea-collar-redesign-forger"}
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control", "family": "authorization", "reason": "", "step_id": "routea-continuum-gap-gatekeeper"}
manuscript_result_elaboration.route_a.certify_scope.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.certify_scope.control", "family": "authorization", "reason": "", "step_id": "certify_scope"}
manuscript_result_elaboration.route_a.formalize_finite_object.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.formalize_finite_object.control", "family": "authorization", "reason": "", "step_id": "formalize_finite_object"}
manuscript_result_elaboration.route_a.derive_proof_packet.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.derive_proof_packet.control", "family": "authorization", "reason": "", "step_id": "derive_proof_packet"}
manuscript_result_elaboration.route_a.audit_prior_art.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.audit_prior_art.control", "family": "authorization", "reason": "", "step_id": "audit_prior_art"}

Semantic ID	Complete recorded detail
manuscript_result_elaboration.route_a.integrate_narrative.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.integrate_narrative.control", "family": "authorization", "reason": "", "step_id": "integrate_narrative"}
manuscript_result_elaboration.route_a.validate_expansion.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.validate_expansion.control", "family": "authorization", "reason": "", "step_id": "validate_expansion"}
manuscript_result_elaboration.route_a.gate_readiness.control	{"disposition": "authorized", "event_id": "manuscript_result_elaboration.route_a.gate_readiness.control", "family": "authorization", "reason": "", "step_id": "gate_readiness"}

E Artifact and receipt ledger

Semantic ID	Complete recorded detail
No artifact and receipt ledger recorded.	

F Specialized projection ledger

Semantic ID	Complete recorded detail
No specialized projection ledger recorded.	

G Causal and workflow graph ledger

Typed edge summary

precedes: 54, contains: 45, supports: 21, refutes: 1, produces: 1, branches: 10, joins: 10, refines: 1

Edge meanings are explicit: precedes is chronology; supports/refutes/constrains are epistemic relations; contains/produces/branches/joins/refines/validates/blocks/hands-off are workflow relations.

G.1 Nodes

Node	Type	Label	Status
manuscript_result_elaboration.route_a.routea_attack.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea_attack.inference	event	routea_attack produces RouteAAttackPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.5b6f606be7f186	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea_open_points.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea_open_points.inference	event	routea_open_points produces RouteAOpenPointPlan: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded

Node	Type	Label	Status
evidence.315ac9b9630548	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea_symbolic_audit.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	event	routea_symbolic_audit produces SymbolicCASAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.15ddea226fb640	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea_integrate_result.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea_integrate_result.inference	event	routea_integrate_result produces RouteAIntegratedResult: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.4038a046d42c12	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea_second_wave.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea_second_wave.inference	event	routea_second_wave produces Integrated PaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.f864c300875269	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	event	routea-geometric-collar-builder produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.8f455a822ffefc	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	event	routea-same-theory-normalization-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.89afe62c827cf9	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	event	routea-geometric-coefficient-extractor produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded

Node	Type	Label	Status
evidence.e57a22c1a0a375	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	event	routea-finite-margin-evaluator produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.2fc65723cf2e10	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	event	routea-margin-endpoint-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.1c29564458a55f	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	event	routea-endpoint-error-budget-certifier produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.4fac32c287b49e	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	event	routea-endpoint-local-integral-calculator rejects ResultCandidatePacket: The endpoint-error audit fails the initial margin, requiring a redesigned collar before the specialist join.	Not recorded
evidence.1952ee704db1f8	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
obligation.0be0f054580524	obligation	A negative certified margin or same-theory mismatch rejects the integrated result.	Not recorded
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	event	routea-same-theory-factorization-hunter produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.fe49659fa93aea	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	event	routea-collar-redesign-forger produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded

Node	Type	Label	Status
evidence.795cc6fc69f022	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	event	routea-continuum-gap-gatekeeper produces ResultCandidatePacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.76f139d78fc9ef	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.certify_scope.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.certify_scope.inference	event	certify_scope produces TheoremScopeDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.29ba0e221db096	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.formalize_finite_object.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.formalize_finite_object.inference	event	formalize_finite_object produces FiniteObjectDefinition: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.ffd73b47728ad7	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.derive_proof_packet.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.derive_proof_packet.inference	event	derive_proof_packet produces ResultsProofPacket: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.deb4ef68f414b5	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.audit_prior_art.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.audit_prior_art.inference	event	audit_prior_art produces PriorArtBoundaryAudit: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.7807d1cab61465	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.integrate_narrative.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.integrate_narrative.inference	event	integrate_narrative produces IntegratedPaperDraft: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.a8bb26ed9ec475	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.validate_expansion.control	event	authorization	Not recorded

Node	Type	Label	Status
manuscript_result_elaboration.route_a.validate_expansion.inference	event	validate_expansion produces ExpansionValidation: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.8da5549746a255	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.route_a.gate_readiness.control	event	authorization	Not recorded
manuscript_result_elaboration.route_a.gate_readiness.inference	event	gate_readiness produces PaperReadinessDecision: Integrate the finite-margin result but block continuum promotion at the common validation tail.	Not recorded
evidence.4f120a9286f066	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
manuscript_result_elaboration.handoff	event	Typed handoff to theorem_proof_delivery	Not recorded
step:routea_attack	step	routea_attack	observed
step:routea_open_points	step	routea_open_points	observed
step:routea_geometric_collar_builder	step	routea_geometric_collar_builder	observed
step:routea_same_theory_normalization_certifier	step	routea_same_theory_normalization_certifier	observed
step:routea_geometric_coefficient_extractor	step	routea_geometric_coefficient_extractor	observed
step:routea_finite_margin_evaluator	step	routea_finite_margin_evaluator	observed
step:routea_margin_endpoint_certifier	step	routea_margin_endpoint_certifier	observed
step:routea_endpoint_error_budget_certifier	step	routea_endpoint_error_budget_certifier	observed
step:routea_endpoint_local_integral_calculator	step	routea_endpoint_local_integral_calculator	observed
step:routea_same_theory_factorization_hunter	step	routea_same_theory_factorization_hunter	observed
step:routea_collar_redesign_forger	step	routea_collar_redesign_forger	observed
step:routea_continuum_gap_gatekeeper	step	routea_continuum_gap_gatekeeper	observed
step:routea_symbolic_audit	step	routea_symbolic_audit	observed
step:routea_integrate_result	step	routea_integrate_result	observed
step:routea_second_wave	step	routea_second_wave	observed
step:certify_scope	step	certify_scope	observed
step:formalize_finite_object	step	formalize_finite_object	observed
step:derive_proof_packet	step	derive_proof_packet	observed
step:audit_prior_art	step	audit_prior_art	observed
step:integrate_narrative	step	integrate_narrative	observed
step:validate_expansion	step	validate_expansion	observed
step:gate_readiness	step	gate_readiness	observed

G.2 Complete typed edge table

Source	Relation	Target	Adapter / detail
manuscript_result_elaboration.route_a.routea_attack.control	precedes	manuscript_result_elaboration.route_a.routea_attack.inference	None

Source	Relation	Target	Adapter / detail
evidence.5b6f606be7f186	supports	manuscript_result_elaboration.route_a.routea_attack.inference	None
manuscript_result_elaboration.route_a.routea_attack.inference	precedes	manuscript_result_elaboration.route_a.routea_open_points.control	None
manuscript_result_elaboration.route_a.routea_open_points.control	precedes	manuscript_result_elaboration.route_a.routea_open_points.inference	None
evidence.315ac9b9630548	supports	manuscript_result_elaboration.route_a.routea_open_points.inference	None
manuscript_result_elaboration.route_a.routea_open_points.inference	precedes	manuscript_result_elaboration.route_a.routea_symbolic_audit.control	None
manuscript_result_elaboration.route_a.routea_symbolic_audit.control	precedes	manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	None
evidence.15ddea226fb640	supports	manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	None
manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	precedes	manuscript_result_elaboration.route_a.routea_integrate_result.control	None
manuscript_result_elaboration.route_a.routea_integrate_result.control	precedes	manuscript_result_elaboration.route_a.routea_integrate_result.inference	None
evidence.4038a046d42c12	supports	manuscript_result_elaboration.route_a.routea_integrate_result.inference	None
manuscript_result_elaboration.route_a.routea_integrate_result.inference	precedes	manuscript_result_elaboration.route_a.routea_second_wave.control	None
manuscript_result_elaboration.route_a.routea_second_wave.control	precedes	manuscript_result_elaboration.route_a.routea_second_wave.inference	None
evidence.f864c300875269	supports	manuscript_result_elaboration.route_a.routea_second_wave.inference	None
manuscript_result_elaboration.route_a.routea_second_wave.inference	precedes	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	None
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	precedes	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	None
evidence.8f455a822ffefc	supports	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	None

Source	Relation	Target	Adapter / detail
manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	precedes	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	None
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	precedes	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	None
evidence.89afe62c827cf9	supports	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	None
manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	precedes	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	None
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	precedes	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	None
evidence.e57a22c1a0a375	supports	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	None
manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	precedes	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	None
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	precedes	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	None
evidence.2fc65723cf2e10	supports	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	None
manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	precedes	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	None
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	precedes	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	None
evidence.1c29564458a55f	supports	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	None
manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	precedes	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	None
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	precedes	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	None
evidence.4fac32c287b49e	supports	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	None
manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	precedes	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	None

Source	Relation	Target	Adapter / detail
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	precedes	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	None
evidence.1952ee704db1f8	refutes	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	None
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	produces	obligation.0be0f054580524	None
manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	precedes	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	None
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	precedes	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	None
evidence.fe49659fa93aea	supports	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	None
manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	precedes	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	None
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	precedes	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	None
evidence.795cc6fc69f022	supports	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	None
manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	precedes	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	None
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	precedes	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	None
evidence.76f139d78fc9ef	supports	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	None
manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	precedes	manuscript_result_elaboration.route_a.certify_scope.control	None
manuscript_result_elaboration.route_a.certify_scope.control	precedes	manuscript_result_elaboration.route_a.certify_scope.inference	None
evidence.29ba0e221db096	supports	manuscript_result_elaboration.route_a.certify_scope.inference	None
manuscript_result_elaboration.route_a.certify_scope.inference	precedes	manuscript_result_elaboration.route_a.formalize_finite_object.control	None

Source	Relation	Target	Adapter / detail
manuscript_result_elaboration.route_a.formalize_finite_object.control	precedes	manuscript_result_elaboration.route_a.formalize_finite_object.inference	None
evidence.ffd73b47728ad7	supports	manuscript_result_elaboration.route_a.formalize_finite_object.inference	None
manuscript_result_elaboration.route_a.formalize_finite_object.inference	precedes	manuscript_result_elaboration.route_a.derive_proof_packet.control	None
manuscript_result_elaboration.route_a.derive_proof_packet.control	precedes	manuscript_result_elaboration.route_a.derive_proof_packet.inference	None
evidence.deb4ef68f414b5	supports	manuscript_result_elaboration.route_a.derive_proof_packet.inference	None
manuscript_result_elaboration.route_a.derive_proof_packet.inference	precedes	manuscript_result_elaboration.route_a.audit_prior_art.control	None
manuscript_result_elaboration.route_a.audit_prior_art.control	precedes	manuscript_result_elaboration.route_a.audit_prior_art.inference	None
evidence.7807d1cab61465	supports	manuscript_result_elaboration.route_a.audit_prior_art.inference	None
manuscript_result_elaboration.route_a.audit_prior_art.inference	precedes	manuscript_result_elaboration.route_a.integrate_narrative.control	None
manuscript_result_elaboration.route_a.integrate_narrative.control	precedes	manuscript_result_elaboration.route_a.integrate_narrative.inference	None
evidence.a8bb26ed9ec475	supports	manuscript_result_elaboration.route_a.integrate_narrative.inference	None
manuscript_result_elaboration.route_a.integrate_narrative.inference	precedes	manuscript_result_elaboration.route_a.validate_expansion.control	None
manuscript_result_elaboration.route_a.validate_expansion.control	precedes	manuscript_result_elaboration.route_a.validate_expansion.inference	None
evidence.8da5549746a255	supports	manuscript_result_elaboration.route_a.validate_expansion.inference	None
manuscript_result_elaboration.route_a.validate_expansion.inference	precedes	manuscript_result_elaboration.route_a.gate_readiness.control	None
manuscript_result_elaboration.route_a.gate_readiness.control	precedes	manuscript_result_elaboration.route_a.gate_readiness.inference	None
evidence.4f120a9286f066	supports	manuscript_result_elaboration.route_a.gate_readiness.inference	None
manuscript_result_elaboration.route_a.gate_readiness.inference	precedes	manuscript_result_elaboration.handoff	None

Source	Relation	Target	Adapter / detail
step:routea_attack	contains	manuscript_result_elaboration.route_a.routea_attack.control	None
step:routea_attack	contains	manuscript_result_elaboration.route_a.routea_attack.inference	None
step:routea_open_points	contains	manuscript_result_elaboration.route_a.routea_open_points.control	None
step:routea_open_points	contains	manuscript_result_elaboration.route_a.routea_open_points.inference	None
step:routea_geometric_collar_builder	contains	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.control	None
step:routea_geometric_collar_builder	contains	manuscript_result_elaboration.route_a.routea-geometric-collar-builder.inference	None
step:routea_same_theory_normalization_certifier	contains	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.control	None
step:routea_same_theory_normalization_certifier	contains	manuscript_result_elaboration.route_a.routea-same-theory-normalization-certifier.inference	None
step:routea_geometric_coefficient_extractor	contains	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.control	None
step:routea_geometric_coefficient_extractor	contains	manuscript_result_elaboration.route_a.routea-geometric-coefficient-extractor.inference	None
step:routea_finite_margin_evaluator	contains	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.control	None
step:routea_finite_margin_evaluator	contains	manuscript_result_elaboration.route_a.routea-finite-margin-evaluator.inference	None
step:routea_margin_endpoint_certifier	contains	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.control	None
step:routea_margin_endpoint_certifier	contains	manuscript_result_elaboration.route_a.routea-margin-endpoint-certifier.inference	None
step:routea_endpoint_error_budget_certifier	contains	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.control	None
step:routea_endpoint_error_budget_certifier	contains	manuscript_result_elaboration.route_a.routea-endpoint-error-budget-certifier.inference	None
step:routea_endpoint_local_integral_calculator	contains	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.control	None

Source	Relation	Target	Adapter / detail
step:routea_endpoint_local_integral_calculator	contains	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	None
step:routea_endpoint_local_integral_calculator	contains	manuscript_result_elaboration.route_a.routea-endpoint-local-integral-calculator.inference	None
step:routea_same_theory_factorization_hunter	contains	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.control	None
step:routea_same_theory_factorization_hunter	contains	manuscript_result_elaboration.route_a.routea-same-theory-factorization-hunter.inference	None
step:routea_collar_redesign_forger	contains	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.control	None
step:routea_collar_redesign_forger	contains	manuscript_result_elaboration.route_a.routea-collar-redesign-forger.inference	None
step:routea_continuum_gap_gatekeeper	contains	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.control	None
step:routea_continuum_gap_gatekeeper	contains	manuscript_result_elaboration.route_a.routea-continuum-gap-gatekeeper.inference	None
step:routea_symbolic_audit	contains	manuscript_result_elaboration.route_a.routea_symbolic_audit.control	None
step:routea_symbolic_audit	contains	manuscript_result_elaboration.route_a.routea_symbolic_audit.inference	None
step:routea_integrate_result	contains	manuscript_result_elaboration.route_a.routea_integrate_result.control	None
step:routea_integrate_result	contains	manuscript_result_elaboration.route_a.routea_integrate_result.inference	None
step:routea_second_wave	contains	manuscript_result_elaboration.route_a.routea_second_wave.control	None
step:routea_second_wave	contains	manuscript_result_elaboration.route_a.routea_second_wave.inference	None
step:certify_scope	contains	manuscript_result_elaboration.route_a.certify_scope.control	None
step:certify_scope	contains	manuscript_result_elaboration.route_a.certify_scope.inference	None
step:formalize_finite_object	contains	manuscript_result_elaboration.route_a.formalize_finite_object.control	None

Source	Relation	Target	Adapter / detail
step:formalize_finite_object	contains	manuscript_result_elaboration.route_a.formalize_finite_object.inference	None
step:derive_proof_packet	contains	manuscript_result_elaboration.route_a.derive_proof_packet.control	None
step:derive_proof_packet	contains	manuscript_result_elaboration.route_a.derive_proof_packet.inference	None
step:audit_prior_art	contains	manuscript_result_elaboration.route_a.audit_prior_art.control	None
step:audit_prior_art	contains	manuscript_result_elaboration.route_a.audit_prior_art.inference	None
step:integrate_narrative	contains	manuscript_result_elaboration.route_a.integrate_narrative.control	None
step:integrate_narrative	contains	manuscript_result_elaboration.route_a.integrate_narrative.inference	None
step:validate_expansion	contains	manuscript_result_elaboration.route_a.validate_expansion.control	None
step:validate_expansion	contains	manuscript_result_elaboration.route_a.validate_expansion.inference	None
step:gate_readiness	contains	manuscript_result_elaboration.route_a.gate_readiness.control	None
step:gate_readiness	contains	manuscript_result_elaboration.route_a.gate_readiness.inference	None
step:routea_attack	precedes	step:routea_open_points	None
step:routea_symbolic_audit	precedes	step:routea_integrate_result	None
step:routea_integrate_result	precedes	step:routea_second_wave	None
step:routea_second_wave	precedes	step:certify_scope	None
step:certify_scope	precedes	step:formalize_finite_object	None
step:formalize_finite_object	precedes	step:derive_proof_packet	None
step:derive_proof_packet	precedes	step:audit_prior_art	None
step:audit_prior_art	precedes	step:integrate_narrative	None
step:integrate_narrative	precedes	step:validate_expansion	None
step:validate_expansion	precedes	step:gate_readiness	None
step:routea_open_points	branches	step:routea_collar_redesign_forger	None
step:routea_collar_redesign_forger	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_continuum_gap_gatekeeper	None
step:routea_continuum_gap_gatekeeper	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_endpoint_error_budget_certifier	None
step:routea_endpoint_error_budget_certifier	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_endpoint_local_integral_calculator	None
step:routea_endpoint_local_integral_calculator	joins	step:routea_symbolic_audit	None

Source	Relation	Target	Adapter / detail
step:routea_open_points	branches	step:routea_finite_margin_evaluator	None
step:routea_finite_margin_evaluator	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_geometric_coefficient_extractor	None
step:routea_geometric_coefficient_extractor	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_geometric_collar_builder	None
step:routea_geometric_collar_builder	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_margin_endpoint_certifier	None
step:routea_margin_endpoint_certifier	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_same_theory_factorization_hunter	None
step:routea_same_theory_factorization_hunter	joins	step:routea_symbolic_audit	None
step:routea_open_points	branches	step:routea_same_theory_normalization_certifier	None
step:routea_same_theory_normalization_certifier	joins	step:routea_symbolic_audit	None
step:routea_symbolic_audit	refines	step:certify_scope	{"from": "ResultCandidatePacket", "to": "RefinedResultCandidates"}

H Projection completeness and omission ledger

Packet field	Disposition	Destination	Reason
alternatives_and_failures	rendered	negative-path body	Not recorded
artifacts_and_receipts	summarized_with_complete_appendix	artifact appendix	Not recorded
authority_ledger	rendered	authority body and appendix	Not recorded
claims	summarized_with_complete_appendix	claim ledger appendix	Not recorded
coverage	rendered	validation dashboard	Not recorded
decisions_and_gates	summarized_with_complete_appendix	decision and gate appendix	Not recorded
declared_route	rendered	workflow route body	Not recorded
evidence	summarized_with_complete_appendix	evidence ledger appendix	Not recorded
graph	summarized_with_complete_appendix	graph summary and complete edge appendix	Not recorded
handoffs	rendered	handoff body	Not recorded
identity	rendered	cover identity	Not recorded
inferences	rendered	expert inference dossiers	Not recorded
lens	rendered	cover and print profile	Not recorded
obligations	rendered	obligation body and appendix	Not recorded
profile_topology	rendered	route topology body	Not recorded
schema_version	rendered	build provenance	Not recorded
scope_ledger	rendered	scope body and appendix	Not recorded
specialized_projections	summarized_with_complete_appendix	specialized projection appendix	Not recorded

Packet field	Disposition	Destination	Reason
timeline	summarized_with_complete_appendix	complete chronology appendix	Not recorded

I Build provenance

Source packet lens

{"accent": "#2fd6b5", "focus_question": "How do specialist results join without crossing finite, endpoint, or continuum boundaries?", "labels": {"claims": "Specialist result packets", "failures": "Identity, endpoint, and margin failures", "handoffs": "Specialist join and validation tail", "scope": "Finite/asymptotic /continuum ledger"}, "lens_id": "manuscript_result_elaboration:route_a", "mandatory_spine": ["routea_attack", "routea_open_points", "routea_integrate_result", "routea_second_wave", "certify_scope", "audit_prior_art", "integrate_narrative", "validate_expansion", "gate_readiness"], "metrics": ["semantic_coverage", "evidence_resolution", "open_obligations", "trace_gaps"], "required_views": ["route", "inferences", "claims", "failures", "scope", "authority", "handoffs", "coverage"], "spine_label": "Specialist fork/join to common validation tail", "visible_name": "Route A Manuscript Attack"}

Packet SHA-256

67046d32dd47f0ee7a76ad2bfc2868284a6fb54f9fa7ed39f3c14e982a058f7f

Renderer version

1.0

Print inventory schema

1.0